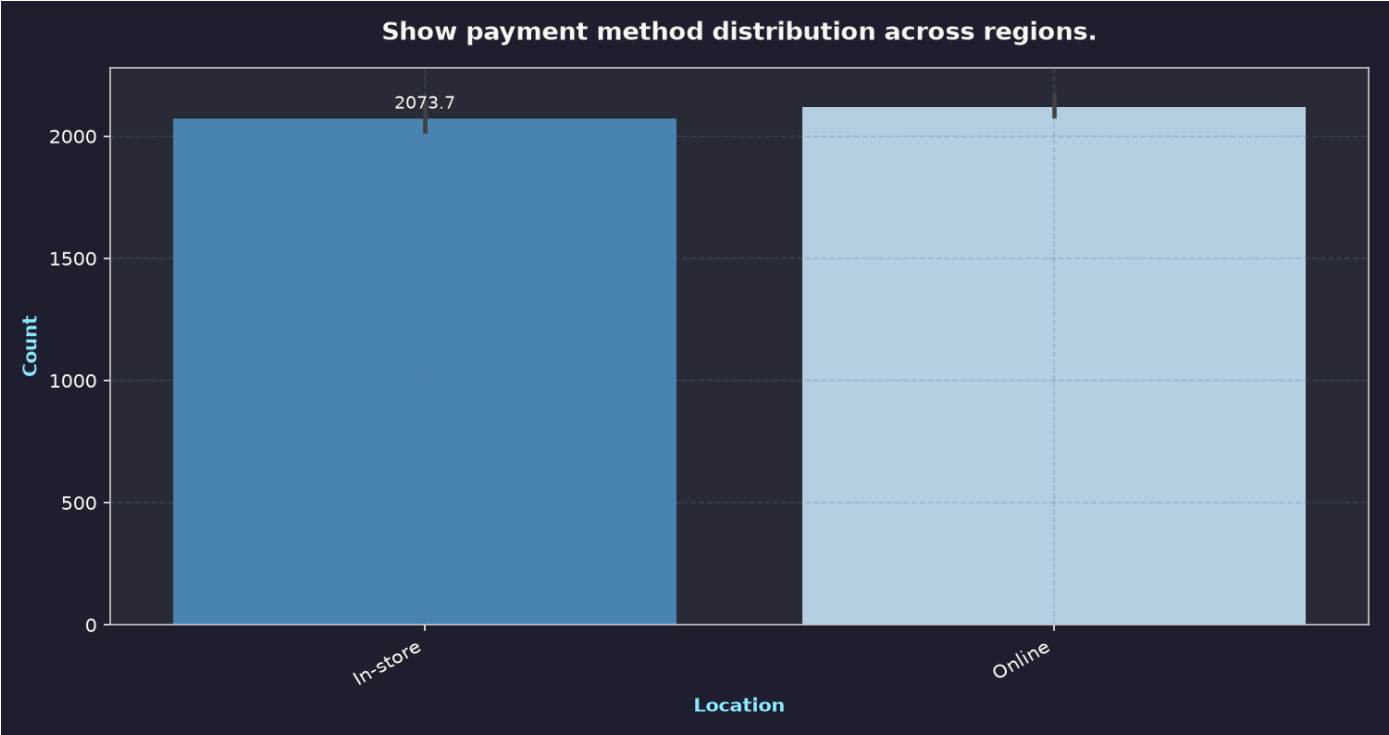


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** “Show payment method distribution across regions.”
- **Answer:** Both **In-store** and **Online** channels exhibit a **balanced mix** of payment methods.
- **Cash** is the most used method in each region ($\approx 34\%$).
- **Digital Wallet** and **Credit Card** follow closely, each accounting for roughly $32\text{--}33\%$ of transactions.
- The distribution suggests no single dominant payment channel, offering flexibility for targeted marketing or operational adjustments.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate rows** | 0 |

| **Missing values** | `Item`, `Price Per Unit`, `Quantity`, `Total Spent`, `Discount Applied` |

| **Potential date columns** | `Transaction Date` (string → datetime) |

| **Key categorical fields** | `Location`, `Payment Method`, `Item`, `Discount Applied` |

| **Numeric fields** | `Price Per Unit`, `Quantity`, `Total Spent` |

3. Data Cleaning & Validation Summary

| **Step** | **Action** |

|-----|-----|

| **Copy & deduplicate** | `result = df.copy(); result = result.drop_duplicates().reset_index(drop=True)` |

| **Trim whitespace** | Stripped all object-type columns. |

| **Date conversion** | `Transaction Date` → `pd.to_datetime(..., errors='coerce')`. |

| **Impute numeric missing values** | Filled with column median (`Price Per Unit`, `Quantity`, `Total Spent`). |

| **Impute categorical missing values** | `Item` → `"Unknown"`. |

| **Impute boolean missing values** | `Discount Applied` → `False` and cast to `bool`. |

| **Validation** | Cleaned DataFrame stored in `result`; all rows retained (12,575). |

4. User Query & Analytical Approach

• **Natural language question:** “Show payment method distribution across regions.”

• **Analytical methodology:**

1. **Group** the data by `Location` (region) and `Payment Method`.
2. **Count** transactions per combination.
3. **Calculate** each method's percentage share within its region.
4. **Sort** for readability.

The implementation uses only Pandas operations (`groupby`, `size`, `reset_index`, `transform`, `sort_values`).

5. Python Analysis Code & Executed Data Output

```
# Count transactions per location and payment method counts = ( df.groupby(['Location',
'Payment Method']) .size() .reset_index(name='Count') ) # Compute percentage within each
location counts['Percent'] = ( counts['Count'] /
counts.groupby('Location')['Count'].transform('sum') * 100 ).round(2) # Sort for readability
result = counts.sort_values(['Location', 'Count'], ascending=[True,
False]).reset_index(drop=True) result
```

Resulting table (**result**)

Location	Payment Method	Count	Percent
In-store	Cash	2,138	34.37
In-store	Digital Wallet	2,063	33.16
In-store	Credit Card	2,020	32.47
Online	Cash	2,172	34.18
Online	Credit Card	2,101	33.07
Online	Digital Wallet	2,081	32.75

(Values are rounded to two decimal places.)

6. Key Insights & Strategic Recommendations

Insights

- **Even distribution:** No payment method dominates; each holds roughly one-third of transactions per region.
- **Cash edge:** Slightly higher cash usage in both channels (≈ 34 %).
- **Regional parity:** The pattern is virtually identical between In-store and Online, indicating consistent customer preferences across purchase channels.

Recommendations

1. **Maintain multi-payment support** – Since customers are split across all three methods, continue offering cash, credit card, and digital wallet options to avoid friction.
2. **Targeted promotions:**
 - **Cash-centric offers** (e.g., small discounts) could modestly boost cash usage further, useful where cash handling costs are lower.
 - **Digital wallet incentives** (e.g., loyalty points) may shift a fraction of the 33 % share toward a faster, lower-cost settlement method.

3. **Monitor trends:** Set up a quarterly dashboard to detect any drift in payment preferences, especially as new fintech solutions emerge.

4. **Operational planning:** Align staffing and POS equipment provisioning proportionally (e.g., ensure enough cash registers and digital wallet scanners) based on the observed ~34 % cash share.

Prepared by the autonomous AI Data Analyst.

![Show payment method distribution across regions. Chart](analysis_chart.png)